

ASSIGNMENT 2

EE673 DIGITAL COMMUNICATION NETWORKS 2025-2026 (Even)

Deadline: 21st Feb 2026

Instructions:

1. Submissions are to be made through the HelloIITK portal
 2. You are required to submit a single zip file containing well commented and easy to understand codes and a single PDF (containing answers to theory questions, if required). There should also be a readme file with clear instructions on how to run the codes.
 3. Kindly name your submission file as <RollNo>_<Name>, eg: 18204269_LavishArora
 4. 10% Marks will be deducted for all submissions that do not follow the above guidelines.
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Question 1: Wireshark Labs

Attempt the TCP and UDP sections of the Wireshark Labs from the reference material Computer Networking A Top-Down Approach 6th Edition by Kurose and Ross (Links are given below). You are required to submit the answers for the questions that are part of the above Labs.

TCP: t.ly/vcH6

[7.5 marks]

UDP: t.ly/K2RS

[2.5 marks]

Question 2: Studying TCP Congestion Control in NetSim

NetSim is a simulation and emulation tool for network design and planning. It also provides tools for protocol and network modeling. Let us follow the preliminary setup before diving into the problem.

1. Install NetSim¹ on your system using the instructions given here², make sure all the sub-modules like Wireshark and winPcap are selected during the installation procedure.
2. Go through the introduction to NetSim (pdf)
3. Go through introduction to TCP using NetSim (pdf and videos)

Once you are done with the preliminaries let us simulate a basic network to study various TCP Congestion Control algorithms. Take the following steps to setup the network

1. Client and Server are connected by a lossy link with the topology shown in the figure 1.
2. An FTP application seeks to transfer large file using TCP
3. Use the configuration in Tab.1 to set the network parameters

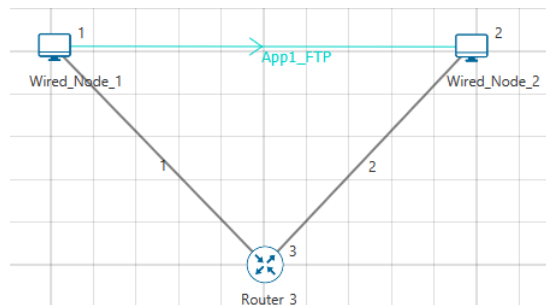


Figure 1: Network topology

¹NetSim is only available for windows, if you are using macOS or Linux you can install and run netsim in a virtual machine.

²Installation instructions: <https://www.iitk.ac.in/nt/faq/netsim.htm>

Table 1: Simulation Parameters

Link Parameters	
Wired link speed	10 Mbps
Wired link BER	1e-7
Wired link propagation delay	25 msecs
Transport Layer Parameters	
Transport Protocol	TCP
Congestion Control	Old_Tahoe, Reno, Cubic
MSS	1460
Application Parameters	
Application	FTP
File Size	1e8 Bytes
Miscellaneous	
Simulation Time	20 secs
Wireshark	Enabled On

After setting up the network run the simulation for different congestion control algorithms mentioned in the table. For each algorithm plot the window size vs time through Wireshark capture. In your answer for this question, display, compare and comment on the three plots. Which Congestion control algorithm is able to utilize the link capacity to its maximum? *[30 marks]*

Question 3: Speed Test using iPerf

iPerf3 is a tool for active measurements of the maximum achievable bandwidth on IP networks. For this question pair up with one of your classmate and measure the bandwidth between your laptops/systems using iPerf (try doing this when you are in different locations). Include screenshot of client and server side outputs. *[10 marks]*

END OF ASSIGNMENT